DATA DICTIONARY

Simulated fill-finish site: data extracts

What this is

A simulated dataset for a fictional contract manufacturing site that fills biologic drugs into vials and prefilled syringes. The site has three aseptic filling lines, shared visual inspection and packaging, QA batch release and dispatch to clients.

- **Period:** batches filled from 1 October 2025 to 1 October 2026 (one year).
- **Extract time:** 5 October 2026, 06:00. Anything after that moment is not in the data, so the most recent batches are still in QA review and some orders are still open.
- **Names:** all products, clients, suppliers and people are fictional.
- **How it was made:** the data was generated with an AI assistant (Claude). Data problems of the kind found in real plant data were added on purpose. They are not listed here, because checking the data is part of the task.

The brief

Site leadership has asked for three things:

1. A standard set of KPI definitions: yield, first-pass quality, scrap, downtime, throughput and on-time delivery, with assumptions and calculation logic.
2. A weekly operations dashboard built on those definitions.
3. The biggest improvement opportunities in the data, with the evidence and a recommended action for each.

How the process works

1. A client places a **sales order** for one to three batches of one product. The site confirms a dispatch date.
2. Each batch has a **process order** in ERP with a planned start and finish.
3. On the filling line, the batch goes through **setup** (line clearance, or a full changeover when the product changes) and then **filling**. Operators log stops of 5 minutes or more in MES.
4. The batch goes to **automated visual inspection** (AVI-1 for vials, AVI-2 for syringes). If the reject rate is above the alert limit of 2.5%, the rejects are **re-inspected** and the final counts are recorded. The action limit is 4.0%.
5. **QC samples** are taken, and the batch is **labelled and packed** (PK-1 for vials, PK-2 for syringes).
6. QA reviews the batch record and **releases or rejects** the batch. A batch cannot be released while a major or critical deviation on it is open.
7. Released batches are **shipped**. Due dates refer to dispatch from the site.

Twice a year each line runs a **media fill** (aseptic process simulation). Media fills use line time but produce no saleable product. They are incubated and inspected outside these systems.

Lines and shifts

Line	Product format	Rated speed	Shift pattern
FL-1	Vials	200 units/min	3 shifts, Monday 06:00 to Saturday 06:00
FL-2	Vials (older line)	120 units/min	3 shifts, Monday 06:00 to Saturday 06:00
FL-3	Prefilled syringes	150 units/min	2 shifts (06:00 to 22:00, weekdays) until 27 February 2026, then 3 shifts

Shifts: Day 06:00 to 14:00, Evening 14:00 to 22:00, Night 22:00 to 06:00. Each line has four rotating crews. The site closes for Christmas (22 December 2025 to 4 January 2026) and for two weeks in summer (13 to 26 July 2026), and on public holidays. `plan_line_schedule` has the scheduled hours per line and day.

File conventions

- CSV, UTF-8, comma separated, decimal point.
- Dates as `YYYY-MM-DD`, timestamps as `YYYY-MM-DD HH:MM`.
- Timestamps in the ERP, MES, QMS and CMMS extracts are in site local time (Copenhagen).
- Empty cells mean the value is missing or not yet known.

Tables

ERP

`erp_clients.csv` (5 rows)

One row per client.

Column	Description
`client_id`	Client key
`client_name`	Client name
`region`	EU or US

`erp_products.csv` (11 rows)

Product master. One row per product, including the two media-fill product codes.

Column	Description
`product_id`	Product key
`product_name`	Product name
`client_id`	Owning client (blank for media fills)
`product_type`	Commercial or Media fill
`container`	Container format
`fill_volume_ml`	Label fill volume
`target_fill_weight_g`	Target fill weight per unit, including overfill
`approved_lines`	Lines approved for the product, separated by ;
`std_cost_eur_per_unit`	Standard cost per filled unit, used to value scrap

`erp_suppliers.csv` (6 rows)

Component suppliers.

Column	Description
`supplier_id`	Supplier key
`supplier_name`	Supplier name
`supplies`	What the supplier delivers
`country`	Country

`erp_sales_orders.csv` (488 rows)

One row per client order. Each order covers one to three batches of one product.

Column	Description
`sales_order_id`	Order key
`client_id`	Client
`product_id`	Product
`order_qty_units`	Ordered units
`order_date`	Date the order was entered
`requested_date`	Dispatch date the client asked for
`first_confirmed_date`	Dispatch date the site first confirmed
`current_due_date`	Due date in ERP at extract time
`due_date_changed_on`	Date of the latest due-date change (blank if never changed)
`due_date_change_reason`	Reason entered with the latest change
`order_status`	Open, Partially shipped, Shipped or Closed - short shipped

`erp_process_orders.csv` (669 rows)

Production orders. One row per planned commercial batch.

Column	Description
`process_order_id`	Process order key
`sales_order_id`	Sales order the batch is made for
`product_id`	Product
`line_id`	Planned filling line
`planned_qty_units`	Planned (theoretical) units
`planned_start`	Planned start of setup
`planned_finish`	Planned end of filling
`po_status`	Released (not started), In process or Completed

`erp_shipments.csv` (637 rows)

One row per batch shipped.

Column	Description
`shipment_id`	Shipment key
`sales_order_id`	Order
`batch_id`	Batch shipped
`ship_date`	Dispatch date
`qty_units`	Units shipped

MES

`mes_batches.csv` (680 rows)

One row per batch, commercial and media fill. units_good = units_filled - units_rejected_inspection - qc_sample_units.

Column	Description
`batch_id`	Batch key
`process_order_id`	ERP process order (media fills use internal orders, IO-...)
`product_id`	Product
`line_id`	Filling line
`batch_type`	Commercial or Media fill
`theoretical_units`	Units the formulated bulk could fill at target volume
`units_filled`	Filled and closed units sent to inspection
`line_loss_units`	Units rejected or lost at the filling line
`units_rejected_inspection`	Units rejected at visual inspection (final, after any re-inspection)
`qc_sample_units`	Units taken for QC testing and retention samples
`units_good`	Units available for release
`reinspection_flag`	Y if the rejects were re-inspected
`fill_start`	Start of filling
`fill_end`	End of filling
`shift_at_fill_start`	Shift when filling started
`crew_at_fill_start`	Crew when filling started
`batch_status`	Status at extract time

`mes_batch_steps.csv` (2,721 rows)

One row per batch and process step.

Column	Description
`batch_id`	Batch
`step`	SETUP, FILLING, INSPECTION, RE-INSPECTION or PACKAGING
`resource_id`	Line or machine used
`step_start`	Step start
`step_end`	Step end (blank if still running)

`mes_inspection_rejects.csv` (4,014 rows)

Inspection rejects per batch and defect type (final counts). Only rows above zero.

Column	Description
`batch_id`	Batch
`defect_code`	Defect code
`defect_description`	Defect description
`units_rejected`	Units rejected for this defect

`mes_component_usage.csv` (1,798 rows)

Primary packaging components issued to each commercial batch.

Column	Description
`batch_id`	Batch
`component_type`	Container, Closure or Cap
`material_id`	Material key
`material_description`	Material
`supplier_id`	Supplier
`supplier_lot`	Supplier lot number
`qty_issued_units`	Units issued

`mes_downtime_events.csv` (1,859 rows)

Line stops logged by operators, 5 minutes or longer. Setup and changeover time is in mes_batch_steps, not here.

Column	Description
`event_id`	Event key
`line_id`	Line
`equipment_id`	Equipment (...-LINE means line level)
`batch_id`	Batch running at the time
`phase`	Setup or Filling
`stop_type`	Planned or Unplanned
`reason_code`	Reason code
`reason_text`	Reason
`start_time`	Stop start
`end_time`	Stop end
`duration_min`	Duration in minutes, calculated by the system
`shift`	Shift at stop start
`work_order_id`	Corrective work order raised for the stop, if any

CMMS

`cmms_equipment.csv` (21 rows)

Equipment master for the lines and the shared machines.

Column	Description
`equipment_id`	Equipment key
`equipment_name`	Name
`line_id`	Line (blank for shared machines)
`equipment_type`	Type code
`install_year`	Year installed
`criticality`	A (critical) or B

`cmms_work_orders.csv` (273 rows)

Maintenance work orders: 4-weekly preventive maintenance (PM), corrective repairs and shutdown work.

Column	Description
`work_order_id`	Work order key
`equipment_id`	Equipment
`wo_type`	PM, Corrective or Shutdown
`description`	Description
`created_date`	Date created
`scheduled_date`	Planned date (PM and shutdown work)
`completed_date`	Date completed
`status`	Completed, Completed late, Cancelled or Open
`deferral_reason`	Reason given when the work was not done on time

QMS

`qms_deviations.csv` (223 rows)

Quality deviations raised against batches.

Column	Description
`deviation_id`	Deviation key
`batch_id`	Batch
`line_id`	Line
`opened_date`	Date opened
`category`	Category
`severity`	Minor, Major or Critical
`status`	Open or Closed
`closed_date`	Date closed
`root_cause_category`	Root cause category, set at closure
`capa_required`	Y if a corrective and preventive action was required

`qms_batch_release.csv` (669 rows)

QA review and disposition per packed batch.

Column	Description
`batch_id`	Batch
`record_received_date`	Date the complete batch record reached QA
`review_started_date`	Date QA started the review
`decision_date`	Date of the release decision
`disposition`	Released, Rejected, Under review or Awaiting review

HISTORIAN

`historian_line_hourly.csv` (26,489 rows)

Hourly readings per filling line from the line historian.

Column	Description
`timestamp`	Start of the hour
`line_id`	Line
`speed_upm_avg`	Average speed over the hour (units per minute)
`units_counted`	Units counted by the filler in the hour
`fill_weight_mean_g`	Mean in-process fill weight
`fill_weight_sd_g`	Standard deviation of fill weight
`room_temp_c`	Filling room temperature
`room_rh_pct`	Filling room relative humidity
`alarm_count`	Machine alarms in the hour

PLANNING

`plan_line_schedule.csv` (1,107 rows)

Scheduled (crewed) hours per line and calendar day.

Column	Description
`date`	Calendar day
`line_id`	Line
`scheduled_hours`	Hours the line was scheduled that day
`shift_pattern`	Shift pattern in force
`non_working_reason`	Why the line was not scheduled, if it was not

`kpi_targets.csv` (180 rows)

Monthly KPI targets set by site leadership.

Column	Description
`month`	Month (YYYY-MM)
`scope`	Site or line
`kpi`	KPI name
`target`	Target value
`unit`	Unit
`direction`	Whether higher or lower is better
`owner`	Owning function
`definition_note`	Definition, where one was given

How the tables connect

From	Key	To
erp_sales_orders	sales_order_id	erp_process_orders, erp_shipments
erp_process_orders	process_order_id	mes_batches
mes_batches	batch_id	mes_batch_steps, mes_inspection_rejects, mes_component_usage, mes_downtime_events, qms_deviations, qms_batch_release, erp_shipments
mes_downtime_events	work_order_id	cmms_work_orders
mes_downtime_events, cmms_work_orders	equipment_id	cmms_equipment
several	product_id, client_id, supplier_id, line_id	erp_products, erp_clients, erp_suppliers, plan_line_schedule, kpi_targets
historian_line_hourly	line_id and time	mes_batch_steps (filling windows)

Where to find the data for each KPI

KPI	Main tables
Yield	mes_batches, qms_batch_release
First-pass quality	mes_batches, qms_deviations
Scrap	mes_batches, mes_inspection_rejects, erp_products (standard cost)
Downtime	mes_downtime_events, mes_batch_steps, plan_line_schedule
Throughput	mes_batches, mes_batch_steps, plan_line_schedule
On-time delivery	erp_sales_orders, erp_shipments

`kpi_targets` has the monthly targets. Most targets come without a definition, so part of the task is deciding what each one should mean.